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ZOO 505 Final Terms Subjective

1. What is ORF? 2

The protein-coding region(s) of each mRNA is composed of a contiguous, non-overlapping string of codons called an open reading frame (commonly known as an ORF).

2. What is RNA splicing? 2

The primary transcripts of intron containing genes must have their introns removed before they can be translated into proteins. The process of intron removal is called RNA Splicing.

3. What is spliceosome? 3

In principle, the reactions could go in the other direction, and indeed this can be forced to happen under special circumstances. The transesterification reactions are mediated by a huge molecular "machine" called the **Spliceosome**. This complex comprises about 150 proteins and five RNAs and is similar in size to a ribosome.

4. Write at least three loops tRNA? 3

The U Loopψ:

It is is so-named because of the characteristic presence of U in theψthe unusual base loop. The modified base is often found within the sequence 5'-TCUCG-3'.

The D Loop:

It takes its name from the characteristic presence of dihydrouridines in the loop.

The Anticodon Loop:

As its name implies, contains the anticodon, a threenucleotide-long sequence that is responsible for recognizing the codon by base pairing with the mRNA.

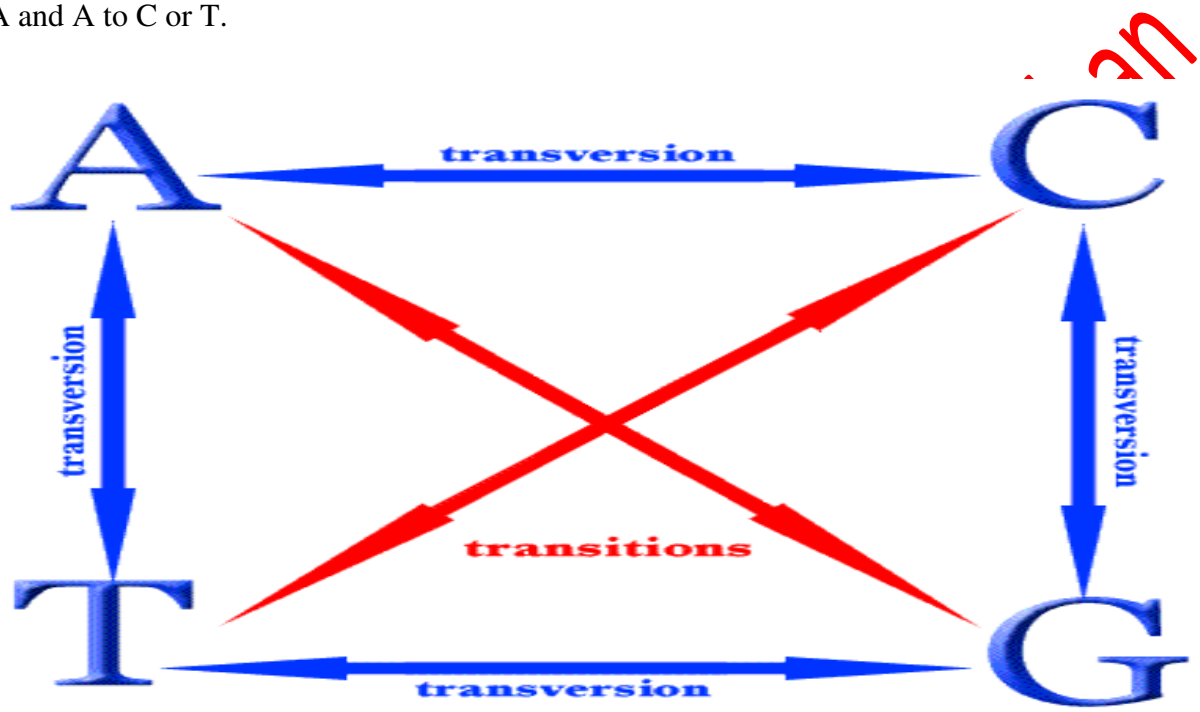
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5. What are transitions and transversions mutation explain with diagram? 5

Transitions are pyrimidine-topyrimidine and purine-topurine substitutions, such as thymine (T) to cytosine (C) and adenine (A) to guanine (G).

Transversions are pyrimidineto-purine and purine-topyrimidine substitutions, such as T to G or A and A to C or T.



6. Write a note on the elongation of transcription? 5

Elongation:

Once the RNA polymerase has synthesized a short stretch of RNA (10 bases), it shifts into the elongation phase.

During elongation, the enzyme performs an impressive range of tasks in addition to the catalysis of RNA synthesis.

1) It unwinds the DNA in front and re-anneals it behind.

2) It dissociates the growing RNA chain from the template as it moves along.

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3) And it performs the proofreading functions.

Recall that during replication, in contrast, several different enzymes are required to catalyze a similar range of functions.

7. Write the elements of Pol II core promoter? 3

The elements found in Pol II core promoter include the TFIIB recognition element (BRE), the TATA element (or box), the initiator (Inr), and the downstream promoter elements (known as DPE, DCE, and MTE).

8. Write the name of polymerases found in Eukaryotes included plants? 2

All eukaryotes have at least three different ones i.e., Pol I, II, and III; and plants also have a Pol IV and a Pol V.

9. Write a detail note on DNA mutation? 10

DNA mutation can be defined as a permanent transmissible change in the genetic material (DNA/RNA). In other words, it is a permanent change in the nucleotide sequence of the genome of an organism. DNA can be easily damaged even under normal physiological conditions. Many different kinds of chemical and physical agents can damage DNA. Some of these agents are endogenous which are produced inside the cells as a result of normal metabolic pathways. While, some others are exogenous agents which comes from the surrounding environment. On one hand, DNA stability is required to ensure that the genetic information may pass accurately from one generation to the next. It is also required for the correct functioning of thousands of genes. On the other hand the genetic variation is needed to drive evolution. If this variation would be lacking, the new species, including humans, would have not arisen. So the life and biodiversity depend on a happy balance between DNA damage (mutation) and its repair.

10. Write the names of three classes of RNA splicing found in the cell? 3

There are total three classes of splicing found in the cells:-

Nuclear pre-mRNA

Group II introns

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Group I introns

11. DNA helicase. 2

Primase activity is dramatically increased when it associates with another protein that acts at the replication fork called DNA Helicase. Therefore a third class of enzymes, called DNA Helicases catalyze the separation of the two strands of duplex DNA at the replication fork.

12. Alternating splicing? 2

Alternative splicing strategy enables a gene to give rise to more than one polypeptide product. These alternative products are called isoforms.

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